H-I-510D
November 6, 1972
SUPERSEDING
Int. Fed. Spec. HH-I-00510C(GSA-FSS)
November 30, 1970, and
Fed. Spec. HH-I-510b
June 20, 1962

FEDERAL SPECIFICATION

INSULATION TAPE, ELECTRICAL, FRICTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electrical insulation friction tape consisting of cotton sheeting that has been impregnated with a tacky adhesive insulating compound. The tape is furnished in approximately 85-foot rolls.

1.2 Classification. Insulation tape shall be one of the following sizes, as specified (see 6.1):

Size A - 3/4-inch wide.
Size B - 1-inch wide.
Size C - 1-1/2 inches wide.
Size D - 2 inches wide.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding Paperboard.
PPP-B-636 - Boxes, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extend specified herein. Unless a specific issues is identified, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

American Society for Testing Materials (ASTM) Standard:

D 69 - Specifications for Friction Tape for General Use for Electrical Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Tariff Order Section, 1616 P. Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Il 60606.)

3. REQUIREMENTS

3.1 The tape shall conform to ASTM D 69 and shall meet the examination requirements of paragraph 4.4. When specified (see 6.1), the tape must pass the ASTM test for discoloration of copper.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examinations and tests. The examinations and tests shall be in accordance with ASTM D 69, and the additional examinations set forth herein.

4.3 Sampling for examinations. All insulation tape of the same width offered for delivery at one time shall be considered a lot for purposes of examination. Random samples of the insulation tape shall be selected from each lot in accordance with MIL-STD-105. The samples, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of rolls of tape, or in the case of preparation for delivery, in units of packed shipping containers. The inspection levels and acceptable quality level (AQL), expressed in defects per 100 units, shall be as follows:

Examination Paragraph	Inspection Level	AQL
4.4.1	I	2.5
4.4.2	S-3	4.0
4.4.3	S-3	2.5
4.4.5	S-2	2.5

4.4 Examination. Each sample selected shall be examined for compliance with the specified requirements. Examinations shall be conducted as specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4, and 4.4.5. If the number of defects in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.1 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be two continuous yards, disregarding the first three turns of each sample roll.

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Examine	Defect
Appearance	Any tear, cut, or break. Lumps, knots, or unevenness to tape. Dirt, foreign material, or imbedded particles. Adhesive not uniform, areas not coated.
Workmanship	Raveling or frayed edges. Splice in tape. Not uniform texture or finish.

4.4.2 Examination of assembly of roll of tape.

Examine	Defect
Appearance	Tape not evenly, neatly, or tightly wound. Any telescoping of roll. Core missing or damaged.

4.4.3 Examination of the end item for defects in dimensions. Samples selected in accordance with paragraph 4.3 shall be inspected for thickness and width as follows:

Examine	Defect
Thickness	Thickness not in accordance with ASTM D 69.
Width	Width not in accordance with ASTM D 69.

4.4.4 Examination of the end item for average length of tape per roll. The sample unit for this examination shall be one roll of tape. A minimum of five rolls shall be randomly selected throughout the lot and measured for length.

Examine	Defect
Length of roll	Length not in accordance with ASTM D 69.

4.4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A. Each roll of tape shall be packaged in a snug-fitting paperboard box conforming to PPP-B-566. Box shall be closed to prevent accidental opening.

5.1.2 Level B. Each roll of tape shall be packaged in a cellophane or plastic wrap with the identification label secured to the outside of the wrap, or each roll shall be packaged in a printed cellophane or plastic wrap. The printed wrap shall contain the information normally required for the label or unit container. The wrap shall be secured to prevent accidental opening.

5.1.3 Level C. Each roll of tape shall be packaged to prevent damage during shipment from supplier to the initial destination.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. 100 rolls of tape shall be packed in a snug-fitting fiberboard box conforming to PPP-B-636, class weather-resistant. Closure shall be in accordance with method II of the appendix to PPP-B-636.

5.2.2 Level B. 100 rolls of tape shall be packed in a snug-fitting fiberboard box conforming to PPP-B-636, class domestic. Closure shall be in accordance with the appendix to PPP-B-636, method I.

5.2.3 Level C. The insulation tape shall be packed in such a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation cost for such supplies. Containers shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the unit package and the shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of tape required (see 1.2).
(c) Whether requirement and test for discoloration of copper by the tape applies (see 3.1).
(d) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

MILITARY CUSTODIANS:

Army - EL
Navy - YD
Air Force - 70

Review activities:

Army - ME
Navy - AS
Air Force - 70

User activities:

Army - AV, WC, AT
Navy - OS

Preparing Activity:

GSA-FSS

Civil Agency Coordinating Activities:

PCD
ACO
AFS
NBS
BPA
DCG
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